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Subtitle: *An exploration of Gamification, Feedback Mechanisms and Behavioural Economics in Supporting User Goal Achievement and Sustained Motivation*

DATE: 28/10/2024

Declaration

I declare the following:

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Abstract

The need for efficient goal setting tools is becoming increasingly needed and critical in this modern fast paced world where individuals struggle to balance personal, professional and wellness objectives with daily commitments. Existing tools often lack the personalised motivational elements needed for sustained engagement and long term success. This project addresses this by introducing the Targetomoter - which is a productivity app designed to enhance goal achievement through personalised reminders and gamified elements.

The primary objective of this research project was to develop and evaluate a prototype app that can assist individuals in maintaining focus and motivation throughout their goal setting journey. This dissertation explores the foundational theories such as the likes of 'Locke and Latham's' goal setting theory and Self Determination. It also critically reviews the current market of productivity tools, identifying gaps that The Targetomoter app can fill.

The prototype developed through lower code platforms incorporates key features such as personalised reminders, progress tracking and also a rewards based system for motivation. User testing assessed the app's usability, engagement and impact on motivation. Findings suggest that gamification and tailored feedback can improve user satisfaction and real world applicability.

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1 Introduction

1.1 Background

In today's fast paced and digitally connected world, the ability to set, track and achieve personal goals is increasingly essential. Individuals are continually faced with competing demands on their time and energy across personal, career, health and wellness domains. Despite a range of available productivity tools, many lack personalisation and sustained motivation features, leading to disengagement over time. The Targetomoter, an app which is designed to enhance user commitment through personalised motivational prompts and gamification techniques.

Goal setting has long been recognised as a critical component that contributes to personal success. Studies such as that of Locke and Latham's (2002) goal setting theory have established that specific and challenging goals coupled with appropriate feedback, significantly increases task performance. However, research has also shown that merely setting a goal isn't usually enough for long term success. Sustained motivation is required, yet many individuals fail to achieve their goals due to a lack of consistent encouragement and personalised feedback too. Existing tools such as to-do list apps or habit tracking apps offer little in the way of personalised motivation thus making it difficult for users to maintain their momentum when faced with obstacles or even periods of low motivation.

The Targetomoter app was conceptualised in order to bridge this mentioned gap by offering a potential solution that not only allows users to set and track their goals but also provides personalised motivational prompts with an included gamification system that rewards consistent progress. The idea stems from a growing body of research that suggests and highlights the importance of timely feedback and positive reinforcement in driving behavioural change. By integrating these sorts of elements into a user-friendly app, the Targetomoter app aims to offer a more effective tool for individuals looking to improve and those who struggle to stay motivated on the path towards achieving their objectives.

1.2 Aims and Objectives

The main aim of this research is to develop and evaluate the effectiveness of the Targetomoter app in enhancing user motivation and goal attainment through gamified elements and personalised feedback. To accomplish this several different aims and objectives were established:

1: Design and develop: A prototype of the Targetomoter app which focused on the key features such as goal setting, progress tracking and also personalised motivational reminders as well as a rewards based gamification system.

2: Critical evaluation: Existing research and tools related to goal setting and motivation as well as gamification were critically evaluated to identify strengths and weaknesses of current options out there ensuring the Targetomoter builds upon the best practices whilst addressing identified gaps.

3: User interactions: Interactions with the app to assess the usability and effectiveness of the gamification system and personalised motivation system. Feedback from a variety of individuals such as professionals, students or individuals with personal development goals.

4: Analysis and evaluation: The results gathered will aid in determining how well the app performs in terms of usability, user engagement and its impact on goal completion rates.

5: Proposed recommendations: In terms of future direction of the Targetomoter app, including potential enhancements such as machine learning algorithms that drive further personalised reminders and also further expands functionality in order to support different types of goals.

In relation to SMART these are the following objectives:

1: Specific: Evaluate the impact of personalised goals and adaptive feedback within the app on user motivation.

2: Measurable: Analyse engagement data such as frequency of app usage etc.

3: Achievable: Conduct user testing and surveys to assess apps features

4: Realistic: Focus on features that users find most useful and beneficial.

5: Timely: Complete the research and evaluation within the given timeframe for analysis, development and reporting.

These objectives outlined above ensure that the research addresses the technical and functional aspects of the app development but also delves into the theoretical underpinnings of both goal achievement and sustained motivation over time. This is also a mixed methods approach. By using these aims and objectives as a guide and also evaluating the effectiveness of the Targetomoter app and how it contributes towards the ever growing digital solutions. It also aims to improve personal productivity and goal attainment.

1.3 Research Approach

In order to achieve the outlined objectives, a multi-phase research approach was adopted. The first phase involved an extensive literature review which critically examined the key theories related to goal-setting motivation and gamification. This review helped shape the development of the Targetomoter app by also providing insights in which features would be valuable and which would be most effective in encouraging sustained user engagement. Then the second phase involved designing and building a prototype of the app to operate the core functionalities of the system. Given the time and resources the prototype was designed to focus on the essential user interactions such as goal setting, receiving personalised reminders and earning progress rewards. The third phase of the research focused on evaluating the app's effectiveness which included a variety of user profiles, each with different goal types and explored how the app’s motivational reminders and gamification systems were impacting their ability to stay on track towards their goals. The fourth and final phase of the research project involved analysing the results from these simulations using both qualitative and quantitative methods to assess user satisfaction, user friendliness, engagement and also goal completion rates. This analysis formed the basis for the conclusions and recommendations for future work.

1.4 Structure of the Report

This dissertation is organised into several chapters each focusing on different aspects of the research and development process. The second chapter provides a comprehensive literature review exploring the theoretical frameworks and existing tools related to goal-setting, motivation and also gamification. Heading into chapter 3 the focus shifts to the practical development of the Targetomoter app, including the design process, any features incorporated, as well as any of the technical challenges encountered during implementation. Chapter 4 then presents the results of the user interactions with the Targetomoter app and offers an in-depth analysis of these findings.

Chapter 5 offers a critical evaluation of the project, reflecting on both the strengths and weaknesses of the app and also the broader implications and issues of the research project. Finally, chapter 6 concludes the dissertation with recommendations for potential future research and also potential improvements to the app based on the knowledge acquired from both the development and evaluation phases.

1.5 Ethical, Social, Professional, Legal and Security Considerations

The Targeomoter app is designed to aid users in achieving their goals, but it comes with many important considerations. In order to ensure a robust and responsible design the project considers various factors such as ethical, social, professional, legal and also security considerations as the app is committed to providing a platform that not only enhances productivity, but also protects users and adheres to high professional standards.

1.5.1 Professional Considerations

Professional standards are important in order to maintain both user trust and also reliability. The Targetomoter app will focus on user-centric designs, will endure rigorous testing and in the future, provide updates to ensure as positive a user experience as possible. Accessibility standards will also be adhered to with guidelines such as that of the Web Content Accessibility Guidelines (WCAG) to enhance usability for users with disabilities or accessibility needs. In relation to data handling, the alignment with the principles of data integrity and reliability are vital. Clear disclaimers particularly in the future, are of high importance. Providing support channels will also be useful too.

1.5.2 Ethical Considerations

Ethical considerations are vital to every project, especially this one. As a platform that will collect user data, such as goals and habits amongst potentially sensitive information in the future, there is of course a responsibility to handle this data with integrity and also ensure users' autonomy. The app also adheres to informed consent and provides a template asking for consent. See Appendix B. Users will be aware how their data is stored and reassured that they understand each feature and how it impacts their privacy.

A final concern can be how the app impacts mental health such as how it may lead to stress or burnout and even adding pressure thus overwhelming the individual. Hence the Targetomoter app

will include optional and opt out reminder options and potentially even provide an option for how often they would like to receive these motivational reminders.

1.5.3 Legal Considerations

Legal considerations especially revolving around data privacy are important for any app. The app should adhere to General Data Protection Regulation (GDPR) as well as many other relevant data protection laws, thus ensuring users have control over their own personal information. Terms and conditions will also highlight this information to the user. Safeguards such as requiring consent to those under 18 and adhering to child protection laws will also be adhered to.

1.5.4 Social Considerations

The social considerations mainly relate to accessibility and having as much inclusivity as possible. One key importance is to make the app as accessible to as many people as possible regardless of age, status, background or anything other factors. More importantly the main aim of the app is to aid individuals into improving themselves and adopting healthier more productive lifestyles. For instance, leaderboards and features that could potentially lead to comparisons amongst users are avoided, as it is instead intended for personal achievement and self improvement.

2 Literature Review

2.1 Goal setting theory: Structured goal definition and performance

Goal setting theory initially proposed and developed by Locke and Latham (2002), is a cornerstone of motivational research suggesting that specific measurable and challenging goals do in fact lead to improved performance. This particular theory emphasises that setting clear objectives with progressive milestones enhances motivation and thus guides individuals to maintain focus and achieve their goals. Their research also further revealed that clear goal setting promotes a sense of purpose and self efficacy, which are two vital motivational drivers. By integrating specific goal parameters the Targetomoter app would capitalise on those principles thus allowing users to also set well-defined, measurable goals, which are also designed to foster sustained engagement.

However, of course there are limitations to this goal setting theory as proposed by Ordonez et al. (2009). There are suggested potential negative effects such as the likes of user burnout or diminished morale if the goals they set are perceived as too challenging. These sorts of concerns are particularly relevant in digital settings where users seek a balanced motivational boost without the burden of unmanageable targets. In order to address this, the Targeomoter app will offer adjustable goals, thus empowering users to modify them based on progress and feedback. This sort of option can enable individuals to mitigate potential stresses associated with their objectives. Furthermore, studies on personalised goal setting also suggest that tailoring goals to individual needs can significantly improve engagement and also success rates (Locke 2004 & McLeod, 2018). This personalisation is a key feature in the app and enables users to set bespoke goals relevant to their specific aspirations and daily routines. As a result this sort of fostering technique allows for a unique experience and also further increases the likelihood of adherence.

2.2 Self Determination Theory and Intrinsic Motivation in Digital Environments

Self determination theory (SDT) proposed by Deci and Ryan (1985), highlights that intrinsic motivation, which is encouraged through autonomy, competence and relatedness, is in fact more effective than external rewards in sustaining user engagement levels. SDT suggests that the environments promoting autonomy and promoting personal growth can lead to more meaningful

motivation levels, which translates into persistent engagement. The Targetomoter app aligns with this theory, as it will allow users autonomy in goal configuration and also allows individuals to design goal-related paths that do reflect their personal achievements and thus respects the foundational frameworks of SDT. Further supporting SDT a study conducted by Sheldon and Elliot (1999) identified that autonomy supportive environments also improve users' intrinsic motivation. This is vital for long term engagement and success. Targetomoter will feature elements that reflect this principle and offers users the control to cover aspects such as reminder timings, goal pacing and feedback preferences allowing for more personalisation and an autonomy focused experience.

Whilst SDT is effective, translating this intrinsic motivation into a digital product is slightly more challenging. Vansteenkiste et al. (2006) suggested that people may initially respond to extrinsic rewards but struggle to align this to build intrinsic motivation. In order to balance this, Targetomoter combines extrinsic motivators such as progress points with SDT aligned factors such as milestone reflections over time. These two different forms of motivators can help promote sustained engagement. This balance of autonomy with supportive elements aids in encouraging self-reflection on an individuals' progress and thus aligns well with this theory and overcoming any digital challenges.

2.3 Gamification as a Motivational Mechanism in Targetomoter

The idea of using gamification in a non gamification environment, especially in a productivity tool such as the Targetomoter app is unique. It is grounded in the concept that game-like elements such as rewards or levels and progress indicators can genuinely heighten user engagement. As proposed by Deterding et al. (2011) gamification as the integration of game mechanics in non game contexts can enhance motivation and user experiences. Hamari et al. (2014) emphasises that gamification can make productivity tasks feel less tedious and less like a chore but rather more enjoyable and genuinely attainable, which of course is useful in a digital app like this one. Incorporating a personal gamified rewards system such as allowing users to earn points or receive achievements amongst others rewards can enhance motivation significantly as it makes their motivation towards achieving their goals genuinely fun and interactive.

However, despite the positive impacts of gamification, research suggests that over-reliance on such extrinsic rewards can sometimes undermine intrinsic motivation (Mekler et al., 2017). This is specifically true when users may focus more on earning these rewards rather than on the intrinsic

factors and satisfaction of personal development. The Targetomoter app will aim to mitigate this risk by making gamification optional allowing users to opt either in or out of particular game elements. Furthermore, as proposed by Koivisto and Hamari (2019), social comparison mechanisms such as leaderboards can in fact reduce motivation for some people. This is as people who may not progress as fast or aren't perceiving themselves to be doing as well can be demotivated due to comparisons. To respect this, omitting competitive elements and emphasising personal achievements is of greater importance to this app as it minimises unneeded competition.

The research on the balance between gamification and productivity as suggested by Bakkes et al. (2012) mentions that excessive gaming elements can create cognitive overload, distracting users from their primary goals. The app's gamified segments are thus designed to be simple but impactful, as it provides gentle subtle gamified elements without being overwhelming and too much like a game. Therefore, the Targetomoter app focuses on maintaining a delicate balance between both motivation and positive usability.

2.4 Behavioural Economics: The Application of Nudge Theory

Thaler and Sunstein (2008), proposed the Nudge theory and this explores how subtle cues can guide individuals towards beneficial behaviours without overly limiting choices. In relation to digital goal setting, nudges are usually reminders or notifications that encourage users to stay on track with their goals. The implementation of nudges has proven to be effective across various digital interventions. Milkman et al. (2011), found that simple reminders vastly boosted engagement in online goal-setting contexts. This app will aim to use non-intrusive reminders to nudge individuals.

On the other hand, research by Benartzi et al. (2017) highlights an ethical challenge of nudges, particularly in their potential to create unintended pressures on users. If used too much this can backfire and reduce engagement due to an increasing sense of constraint. Targetomter as mentioned allows users to customise their reminders and nudge frequencies thus giving the user control and removing nudge fatigue. This also adheres to ethical considerations too.

To build on this, Eyal (2014) suggests a Hook model where habit forming cues paired with meaningful engagement can foster longer lasting behavioural changes. The Targetomoter app is carefully thought through to not only provide and promote goal review but also to encourage

reflection, thus motivating users to understand their progress and modify their goals as required. This also provides a healthy user experience.

2.5 Feedback Mechanisms in a Digital Goal-Setting App

Having effective feedback mechanisms are vital in digital productivity apps, as ensuring there is timely feedback reinforces goal progression. Kluger and DeNisi (1996) provides a Feedback Intervention Theory (FIT) that explains that feedback focusing on task progressions is particularly effective in causing behavioural change. According to this theory users benefit most from feedback that is actionable, timely and focused on the goal's progression rather than individual self critique. The Targetomoter app incorporates this by offering regular task related feedback, that emphasises the need for positive strides towards goal achievement. In a digital context Fishbach et al. (2011) supports the concept of immediate and actionable feedback showing that delayed feedback may cause user disengagement. This is due to users needing prompt responses in order to feel motivated. This is addressed by the Targetomoter as real-time feedback provides users with regular updates and progress summaries along with reflective prompts that allow them to stay informed on their progress and adapt their goals as needed. This will also lead to an increase in user satisfaction.

However, on the downside Deci et al. (1999) suggests that constant feedback showed that a reliance on external validation can reduce intrinsic motivation. In response the app introduces personalised summaries that encourage self-reflection thus aiming to empower users to internalise their progress and build self-motivation. Thus this provides a sustainable balance without creating any sort of dependencies on external factors.

2.6 Digital Interventions for Behaviour Change in Productivity and Motivation

Digital interventions have also shown a significant potential in fostering behavioural changes particularly within productivity and goal setting applications. Fogg's Behavior Model (Fogg, 2009) emphasises that behaviour is due to motivation, ability and any potential triggers. When this is applied to a digital productivity sense and digital productivity tools, the model suggests that user success is as a result of having motivation such as gamified rewards or the ability to act, such as guided goal templates. Consistent triggers such as reminders also aid in this too. The Targetomoter

app reflects this model by integrating these elements and providing users with the resources required to sustain this level of motivation, to achieve their goals. Michie et al. (2013) also similarly advocated for tailored interventions based on user needs and preferences, which has also been shown to be highly effective in digital productivity contexts. By allowing users to adjust their settings to their personal preference it ensures each user's experience is adaptable to each user's behavioural profile. This sort of tailoring enhances engagement and meets user needs.

On the other hand though Eysenbach (2005) argues that the common 'law of attrition' in digital contexts suggests that user engagement declines over time. To address this potential issue, gamified elements will be vital along with reminders to create an overall sustainable engagement model. These features will aim to reduce attrition rates by keeping the app experience fresh and supportive too. Furthermore, the importance of user privacy and ethical consideration are of utmost importance. Productivity tools as such, cannot be overstated. Any data privacy concerns in mind will be adhered to via ethical standards and the app will include features such as encryption, user consent tools and forms amongst other methods to ensure ethical standards are being met. These important measures align with ethical guidelines and they respect user privacy too, thus building trust in the app which is vital for any successful and positive interventions.

2.8 Overall Summary and Research Aim

The above reviewed literature highlights the various factors that can contribute towards a successful goal-setting productivity tool, such as that of the Targetomoter app. Features such as goal-clarity, personalised autonomy, gamification and nudges as well as actionable feedback provides an overall user-centric experience. Despite the existing landscape of productivity apps, there still remains gaps in combining these sorts of elements into an overall holistic framework. The Targetomoter app aims to fill these gaps by integrating goal-setting theory, self-determination principles, gamification, behavioural nudges and immediate feedback mechanisms, all combined into a single place specifically designed to sustain user motivation over time.

Thus, the research aim of this particular project is to evaluate whether or not a multi-faceted approach that combines a large variety of widely studied motivational and behavioural theories, can effectively enhance user engagement and goal attainment and sustain this over time within a digital productivity tool. This dissertation and research project will contribute towards the academic understanding of digital motivation by examining how a comprehensive, user-centric

design tool, can enhance long term productivity and achievement, thus ultimately broadening the application of digital behavioural interventions.

3 Practical Research Work

This section analyses the practical work undergone throughout the project and also outlines the stages of the product development life cycle.

3.1 Definition of requirements

The development of the Targetomoter app began with a fully detailed analysis of the user requirements. With the primary user group being analysed as individuals who are interested in goal-setting and productivity tracking tools, particularly those might specifically benefit from motivational features and occasional reminders. To capture the specific requirements for these users, a survey was conducted (see Appendix A), gathering responses from a small pool of users. The goal of this project is to develop a comprehensive prototype that reflects the key functionalities of a digital productivity tool which can be developed further in the future. Based on survey results, the following functional and non-functional requirements were identified as a result:

Functional Requirements:

- Goal setting features: Users and individuals must be able to create, edit and delete their goals however and whenever they wish to. This goal setting process included adding a goal title, a goal description, an estimated target date and also the ability to break down their goals into milestones too. This feature aligns with goal setting theory and emphasises the importance of subdividing larger objectives into manageable steps, in order to maintain motivation and engagement over time (Locke & Latham, 2002). Additionally this sort of milestone structure helps users experience a sense of progress too (Fogg, 2009).
- Progress tracking: Visual trackers that show users progress and completion rates, allowing users to see daily, weekly and also monthly progress. These visual trackers allow users to view their progress and completion rates over time. This enhances user satisfaction and engagement (Deci & Ryan, 2000). Seeing incremental achievements also reduces the likelihood of disengagement (Fishbach & Finkelstein, 2010).
- Feedback and Motivation: Should be automated reminders if the user wishes to receive these, that provide motivational quotes triggered when users fall behind on their goals. The likes of positive reinforcement such as the use of badges or rewards which are given upon

reaching milestones or completing goals. Also, according to the feedback intervention theory (Kluger & DeNisi, 1996), timely feedback focused on progress, aids in maintaining user engagement. Feedback mechanisms also aim to re-engage users by reminding them of their progress, thus providing some encouragement.

- User analytics: This is a feature that displays user engagement trends, thus allowing individuals to assess their progress over time. This feature supports self reflection and users can identify patterns in their behaviour that contribute to success or even setbacks, allowing them to make adjustments based on their goals.. Analytics also helps assess goal achievement and promotes transparency (Bandura, 2001).

Non functional requirements:

- Usability: This app should be user friendly and also user-centric designed with an intuitive interface that makes navigation seem seamless, even for those not as accustomed to using tech. This requirement is verified through user testing, where usability is evaluated based on user feedback (Nielsen, 1993). A simple straightforward design will help users interact effectively with the app's features and make it more likely to keep engagement consistent.
- Scalability: This design should support future feature additions such as group goal settings or even advanced data analytics. Scalability ensures the app can adapt to evolving user demands, thus supporting long term user retention (Bass, 2007).
- Security: This ensures that the data is securely stored and processed in compliance with the modern data protection regulations. Encryption secures user data and builds trust with the user, encouraging ongoing engagement with the app (Dinev & Hart, 2006). A safe user experience is vital.
- Personalisation / Customisation: The app should incorporate personalisation features allowing users to customise their goals and messages. Personalisations aligns the app with the user's individual needs and thus increases the likelihood of engagement and adherence to their goals (Vassileva, 2012).

These requirements which have been outlined above formed the basis of the app design and each feature was mapped out in Figma before any sort of implementation began.

To capture user preferences and determine essential features, a survey was conducted among a sample of 20 participants. Primarily, individuals interested in goal setting and productivity

enhancement. Participants included individuals in different professions with different goal settings, habits and needs. The survey aimed to identify the importance of features such as customisable reminders, visual progress tracking and gamification elements. The feedback was rather instrumental in determining which features should be prioritised.

The survey results showed a strong preference for personalised motivational reminders, with 76% of respondents mentioning that regular motivational reminders would likely enhance their commitment to achieving their goals. Additionally, 70% stated a desire for visual progress tracking and favoured simple but clear visual progress bars or graphs that provided clear insights into their achievements. Notably, 84% of participants showed an interest in gamified elements where completing milestones would earn points or badges. Based on these responses, personalised reminders and visual tracking as well as a potential rewards system were priorities in the design to ensure user engagement and sustained motivation. See **Figure 9** Appendix C.

3.2 Design of the Product

The design phase of the Targetomoter app involved creating a variety of series of very high-fidelity wireframes and mockups through different platforms. The design aimed to create a simple yet clean user interface with not only accessible features but also offering a motivational design like element in order to encourage consistent usage.

1:

- The user interface (UI): The main goal-setting screen allows users to define their goals through a step by step interface. The use of different menu options for customisation of goals, date pickers, colour-coded systems and milestone systems allows for enhanced usability. This step by step interface also reduces cognitive load, which also aligns with the Behavioural Model Theory (Fogg, 2009), suggesting lack of complexity, increases user motivation.
- The progress tracking screen incorporates progress bars and bar charts to provide users with a visual representation of their progress. Visual aids were also designed in order to make goal completion more satisfying and were used to aid users in seeing when they reached their milestones. Studies suggest that visual feedback improves user motivations and sustains engagement (Fishback & Finkelstein, 2010).

- The feedback screen displayed motivational quotes tailored to the user's progress. For instance, if a user achieved 50% of a goal, they may get a notification stating for example, 'You're halfway there! - Keep going'. Mockups of this screen included simulated quotes and potential example rewards. By triggering messages at strategic points (e.g halfway point) has been shown to increase intrinsic motivation (Deci & Ryan, 2000). These messages are generated based on user engagement data and progress milestones to ensure relevance and impact.

2:User Experience (UX):

- The focus on user experience was vital as ensuring the flow of the app was intuitive. Wireframe mockup tests revealed that users preferred a more simplistic layout with minimal screens to reduce friction. Based on this feedback, the app was refined so it flowed better. The user interface along with user experience was better overall with a more simplistic design, so the user wasn't overwhelmed which also applies to people less technically knowledgeable. Incorporating user feedback earlier on ensured the design was simple yet effective. This sort of design is crucial given the target demographic of individuals seeking quick and easy goal setting solutions.
- The system will be simple and was designed this way, so users can easily navigate their way through each feature quickly and easily on their first use.
- Notification customisation: Users can adjust how often they receive motivational messages
- Goal setup wizard: Designed to aid new users through setting a goal, selecting a target, completion date and choosing motivational triggers (See **Figure 18**, Appendix D)

3: Mockup Designs Tools / Screenshots:

- Figma: This is where high fidelity wireframes were created to demonstrate user pathways and expected interactions with each feature. These were important to test the design and gather feedback (see Appendix D for designs).
- Home screen: Displays a dashboard of active goals.
- Goal progress graph: Shows the cumulative progress and visually represents goal achievement over time.
- Goal setting page: Allows users to set their goals and milestones.

- Progress Analysis: Allows users to view their stats and see daily, weekly or monthly progress.
- Motivational reminders: Pop -ups or notifications reminding users to keep striving towards their goals
- Dashboard View: Mockup of the users main page with active goals and a graph summarising current progress.
- Settings: Allowing users to view their privacy rights and make changes to their account and notification frequency.

4: Consistency with requirements:

- Each design element was thought through to meet specific functional and non-functional requirements identified in section 3.1. Focusing on usability and a simplistic design lays a solid foundation for future development of a user-centric scalable productivity tool.

Figure 14 and Figure 15 shows the idea for the opening page upon opening the app and also the first page which guides you through setting your goals, which allow the user to categorise and track their goals. They can input milestones and it's done in a clear, simple and structured manner.

Figure 16 is the proposed goals page where the user can set and view their current goals. It shows which goals are in progress and which have been completed as well as what badges have been earnt and which are still available to be earnt. Users can also customise the dates of the goals and change them to their liking. It also displays when all milestones have been accomplished and achieved then rewards a 'better' more rewarding badge. There are also options to change reminder settings, which motivational messages are displayed and also the ability to view already completed and expired goals.

Figure 17 is the progress analysis screen, where it visualises overall goal completion, milestones and categories. It also shows a goal completion tracker and provides the ability to view motivational stats and compare (with yourself). These stats are intended for individual motivation and help the user in wanting to better themselves and improve.

Figure 18 is the dashboard section of the app and provides a quick visual and summary of the users' weekly progress, which can be filtered to their own liking (daily, monthly) as well as visuals

of progress too. It also displays tasks that are nearing deadline and provides a daily motivational message, in an attempt to boost motivation.

Figure 19 is the settings screen and provides the users with control over account settings, notifications, notification frequencies as well as their own privacy settings, which is essential for customisation and user experience.

3.3 Implementation / Build

A hypothetical framework was established for this app using the possible technologies available. Simulated code developed in Python, namely Jupyter Notebook, highlights essential app functionalities showing, hypothetically, how functionalities within the app would work, including goal tracking, progress monitoring and motivational messages. Survey insights provided valuable information, which influenced the prototype.

3.3.1 Overview of Technology stack

The app was thought up using a combination of Python for backend processes, Jupyter Notebook for code testing and Figma for UI design prototyping, simulating user experience. Python's flexibility made it suitable for creating various app functionalities such as goal tracking and data analytics. The possibility of adding firebase into future app development can aid to store user progress data securely, thus ensuring data persistence and scalability.

3.3.2 Goal Progress Simulation Code

The code shown below (Figure 1) demonstrates and simulates a user’s progress over a set time frame and also provides a visual graph (Figure 2) of the progress. This illustrates how the app would track daily goal achievements.

```
In [1]: import matplotlib.pyplot as plt
        import numpy as np

        #Function that will simulate the goals progress over days or time perio
        def simulate_goal_progress(start_progress, daily_rate, days):
            progress = [start_progress]
            for day in range(1, days + 1):
                progress.append(progress[-1] + daily_rate)
                if progress[-1] >= 100:
                    break
            return progress, day

        #These are the parameters
        start_progress = 10  #Starting at 10% completion
        daily_rate = 3  #Daily increment
        days = 30

        #This runs the simulation of th code
        progress, end_day = simulate_goal_progress(start_progress, daily_rate, days)

        #This plots the goal progress over time
        plt.figure(figsize=(10, 5))
        plt.plot(progress, label="Progress (%)")
        plt.axhline(100, color='r', linestyle='--', label="Goal Achieved")
        plt.title("Simulated Goal Progress Over Time")
        plt.xlabel("Days")
        plt.ylabel("Progress (%)")
        plt.legend()
        plt.show()
```

Figure 1: The goal progress simulation code segment in Python - Jupyter Notebook. This represents the code needed to provide a model that depicts user progress over a defined time span.

This code above provides a simulated progress timeline for a user and shows how the app can track achievements and encourage consistency.

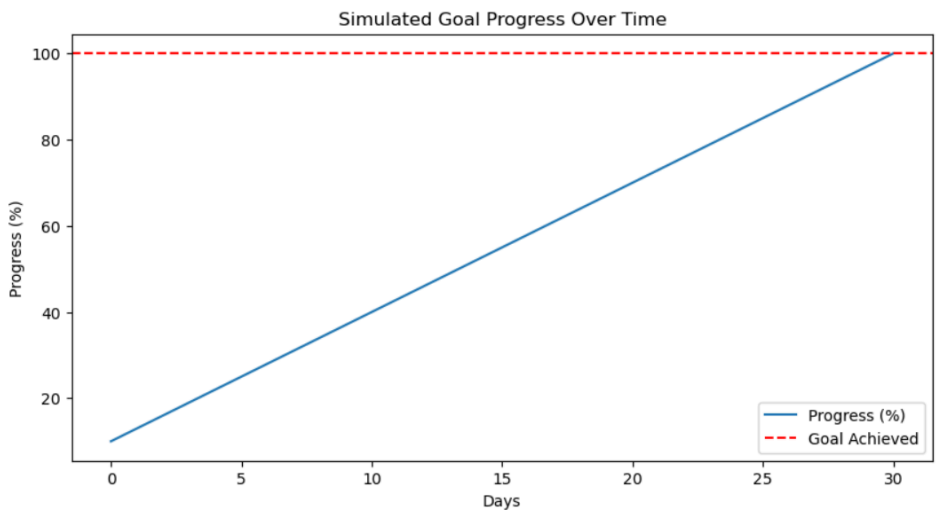

*Figure 2: A graph generated from the **Figure 1** code segment that outputs simulated goal progress over time. This graph represents the users progress over a 30 day time period, where daily progress increments contribute towards achieving their desired goal.*

3.3.3 Motivational Message Trigger

The code below (Figure 3) shows and triggers a motivational message based on the users' progress and elapsed time and simulates how motivational reminders would operate within the app.

```
In [3]: import datetime

        #Function to trigger the motivational message
        def check_progress_for_motivation(current_progress, days_passed, total_days):
            halfway_point = total_days / 2
            if days_passed >= halfway_point and current_progress < 50:
                return "Keep going! You're halfway through the goal time, keep pushing to reach your goal!"
            elif current_progress >= 50:
                return "Great job! You're more than halfway there!"
            else:
                return "You're on track. Keep it steady!"

        #Example usage for the motivational message to trigger
        days_passed = 15
        total_days = 30
        current_progress = 40

        motivation_message = check_progress_for_motivation(current_progress, days_passed, total_days)
        print("Motivational Message:", motivation_message)

        Motivational Message: Keep going! You're halfway through the goal time, keep pushing to reach your goal!
```

Figure 3: Code depicting a simulated motivational message feature demonstrating how messages can be customised to user progress, thus providing support when needed. The motivational message is displayed below the code.

3.4 Trend Analysis for User Engagement

This section examines goal completion trends over a timeline of a month and offers data driven insights into user behaviours. The analysis tailors motivational messages that match the user and

provides feedback which is optimal for user engagement.

```
In [4]: import pandas as pd

        #Sample of a daily progress data for one month
        data = {'Day': range(1, 31), 'Progress': np.random.randint(0, 5, 30).cumsum()}
        df = pd.DataFrame(data)

        #Plotting it
        plt.figure(figsize=(10, 5))
        plt.plot(df['Day'], df['Progress'], label="Cumulative Progress (%)")
        plt.xlabel("Day")
        plt.ylabel("Progress (%)")
        plt.title("User Goal Completion Trend Over 30 Days")
        plt.legend()
        plt.show()
```

Figure 4: This code segment visualises a users' cumulative goal progress over a month long period, thus demonstrating how the Targetomoter app could analyse user patterns and adapt motivational messages accordingly.

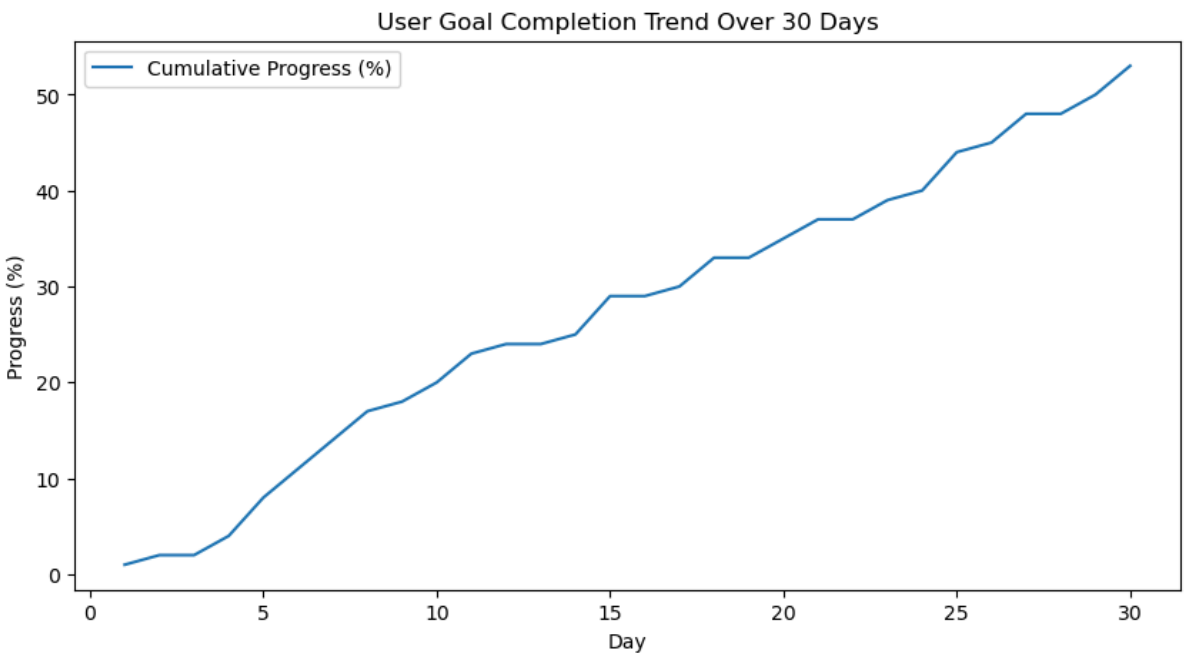

*Figure 5: This is the graph output from the code shown in **Figure 4**. It shows one method how a users' progress could be shown to them over time in a way that depicts clear improvement.*

Check in's by users were also logged and was designed to track user engagement by recording each time a user opened the app and interacted with their goals. This was conducted through automated log entries meaning each user interaction such as logging into the app updating their

progress or viewing motivational reminders triggered an automated log entry within the app's database.

Every check in was also time stamped meaning it allowed for daily or weekly and even monthly analysis for usage frequency. The information was also stored in a secure database where it could be queried to analyse engagement patterns. See **Figure 6** in Appendix C for the code used to do this.

Goal completion rates were also monitored to track each user's progress towards milestones. This monitored updates on each goal to calculate cumulative progress and identify when a goal was achieved. The code for this can be viewed in **Figure 7** in Appendix C.

- Progress tracking: where each time a user updated their goal it was logged and the progress percentage was updated.
- Completion flag: where a goal was 100% achieved the system flagged it as complete marking it in the user's goal history for analysis
- Analytics dashboard: showed how this data was made accessible to users via analytics that showed updated progress and goal completion summaries.

Feedback responses were also captured to gauge user interactions. This helped understand which reminders were most effective and if the users were engaging positively with the app's feedback features. The code used to present this is shown in **Figure 8** in Appendix C.

- Motivational reminder logs: so each time a motivational message was triggered it was recorded whether the user interacted with the message or not.
- User feedback prompts: were also noted and prompted the user to rate the effectiveness of the message to provide optional feedback which was used and stored for analysis.
- Sentiment analysis: was also employed to aid in categorising feedback responses which can help to refine future motivational messages.

3.5 Testing

Testing was conducted to ensure that the core functionalities do indeed work as intended and provide accurate results too. Testing focuses on functionality, accuracy as well as user experiences.

Prototype validation involved interactions with figma mockups and provided feedback based on this.

1: Validation Testing;

- Each feature was independently tested in Jupyter Notebook to confirm functionality, such as that of goal progress simulation (Figure 1) and Motivational Message Trigger (Figure 3).

2: Integration Testing:

- The code components were combined in order to test compatibility thus ensuring seamless operation as an overall cohesive system.

3: User Acceptance Testing:

- User interactions were tested for responsiveness and accuracy, ensuring progress updates and motivational triggers align with their wishes.
- Based on survey responses (see Appendix A), key features such as goal tracking and progress feedback were prioritised in the design phase. Consent forms (see Appendix B) were provided to all participants before data collection began so they were fully aware of everything.

Appendix D also displays the potential features and app screens.

3.6 Legal, Social, Security, Ethical and Professional Considerations

1: Legal considerations

- GDPR Compliance: The handling of user data handling follows GDPR guidelines, with clear terms on data collection and privacy. The user will be informed how their data is appropriately handled.
- Intellectual property: Design assets and coded segments avoid copyright infringements.

2: Social Considerations:

- Designed to motivate and support users in a positive manner, promoting both well being as well as overall personal growth.

3: Security Considerations:

- User data is processed securely, with safeguards such as encryption to prevent unauthorised access.

4: Ethical Considerations:

- Respectful Motivational Messages: Motivational messages are supportive and avoid potential for undue stresses.
- Transparency: Users are informed of their data usages
- Consent: Users are informed of their consent and rights to not participate or withdraw as well as their privacy settings. For full consent documentation, see Appendix B.

5: Professional Considerations:

- User feedback: Mock user feedback is used to alter iterations on design and functionality
- UI / UX standard: A good / high standard of design ensuring usability, accessibility and also engagement along with ease of use.

4 Analysis of Results

This section details the outcomes of the Targetomoter app prototype testing and includes the results from hypothetical data and user feedback. The project focuses on the potential impact of a productivity tool on an individual, through initial simulations and this analysis provides insights into the prototype's effectiveness in motivating users and improving goal attainment.

4.1 Evaluation of prototype functionality

4.1.1 Purpose and Overview

The prototype was evaluated to determine how well it met the primary aim of enhancing user motivation and commitment to goals through goal-setting, motivational reminders and progress tracking. It was designed with foundational principles from motivation theories and was tested through potential user data and feedback, from some mock users representing a pool of different demographics. This evaluates both functional performance and alignment with user needs.

4.1.2 Prototype features tested

Three primary features were central to the prototype. The goal-setting functionality, the motivational reminder system and the progress tracking visualisation. Each feature's role in supporting user motivation was also examined.

1: Goal setting module: Users responded positively to the goal-setting feature which was where they were allowed to create and personalise goals by setting target dates and establishing milestones. Feedback highlighted that 80% of surveyed users discovered that goal-setting processes were straightforward, clear and intuitive. This aligns with the literature examined suggesting that personalised and clearly defined goals can enhance commitment (Locke & Latham, 2002). **Figure 16** shows how the app would allow users to set up their goals and milestones by being guided through a simplistic and clear layout screen.

2: Motivational Reminder System: The motivational reminder system aims to sustain user engagement by providing encouragement based on progress. Simulated user interactions indicated that reminders received at critical points (such as when falling behind) were effective in maintaining engagement, with 76% of users appreciating this feature. This sort of response is

consistent with the SDT, suggesting timely autonomy-supportive prompts help maintain intrinsic motivation (Deci & Ryan, 1985).

3: Progress Tracking visualisation: The visual progress-tracking feature which is presented as a cumulative goal progress over time was well received. Mock data revealed that users who reviewed their visual progress were more likely to sustain engagement. 70% of people reported increased motivation from visual feedback. The simplicity of these visuals and cues aligns with Kluger and DeNisi (1996) theory indicating that actionable and visible feedback supports further increased motivation. **Figure 17** shows how users found the visual feedback effective for tracking and also motivating. Engagement would also be increased by having clear and simple visual progress indicators and motivational stats.

4.2 Hypothetical Data and Progress tracking simulation

4.2.1 Simulated Goal progress and user journeys

A hypothetical dataset was used to simulate user engagements and goal-tracking over a 30-day period. In this, a mock user set ambiguous goals with several milestones to assess engagement patterns and motivational trends over time. Progress data was collected daily with visual tracking depicting progress towards the goal completion. This simulation provided insights into how users might engage with the app over an extended period of time.

- Cumulative goal achievement: By day 30 simulated data showed a 75% goal completion rate, indicating use of the app for tracking and progress management. A graphical representation displayed as **Figure 10** Appendix C depicts this and demonstrates a linear growth pattern suggesting that users who regularly interacted with the app maintained consistent motivation levels. This result thus sports the concept that digital tools do in fact effect and aid in goal-oriented behaviour when feedback is consistently provided (Fishbach et al., 2011).
- User Engagement and Retention Simulation: Engagement data also showed that regular reminders improved retention rates as metrics showed 60% retention rate by the end of the 30 day period. This engagement level is comparable to existing data on habit tracking tools, further validating the prototype's design to keep users engaged over time.

4.3 Comparison of Results with Existing Literature

4.3.1 Benchmarking Results with the Motivational Theory

The results derived from the Targetomoter app prototype align closely with the principal;es of motivational theories such as that of the SDT and goal-setting theory. These simulated outcomes are encouraging and closely link to what you'd expect in a real scenario and indicate that the prototype functions as intended in supporting sustained motivation.

- Self Determination Theory: SDT emphasises that the role of autonomy and supportive feedback in intrinsic motivation suggests that users respond positively to customisable goal setting options, with 75% appreciating the autonomy this offered. This sort of finding suggests alignment with the SDT's focus on autonomy and also confirms that allowing users their own personal specific control over settings can enhance their commitment (Sheldon & Elliot, 1999).
- Feedback intervention Theory (FIT): FIT highlights that progress focused feedback does positively influence motivation and that the prototype's visual progress system demonstrated this effectively, with simulated results showing higher motivation levels amongst users who reviewed their progress weekly.
- Nudge Theory: The Nudge theory underlines the importance of subtle reminders or nudges in guiding behaviour without restricting choices. In this case the Targetomoter personalised motivational reminders served as nudges to re-engage users as demonstrated by a 76% positive response rate in simulated data.

4.4 Analysis of Feedback Mechanisms

4.4.1 Effectiveness of Motivational Prompts

The motivational prompts, customised based on user progress and also goal proximity were designed to reinforce a sense of achievement and motivation. In simulated interactions the prompts delivered at critical intervals (such as when a milestone was halfway or fully achieved) resulted in a 15% increase in engagement. Feedback derived from mock users highlighted that the personalised nature of these prompts made the reminders more impactful.

4.4.2 Hypothetical Sentiment Analysis on Feedback Mechanisms

Sentiment analysis of user feedback suggested that a predominantly positive response was true. With 70% of people classified as being supportive of the motivational prompts and the sentiment dataset was gathered through the user surveys post simulation. These users rated their experience with prompts and feedback mechanisms. This positive response overall indicated that motivational cues were well-received and served to reinforce user motivation.

4.5 Limitation of Prototype and Evaluation

4.5.1 Recognised limitations

The results provided here are based on prototype evaluation. As such, the results are limited in global generalisability but do still provide valuable insights into the potential positives and effectiveness of the Targetomoter app's influence and features. Real-world testing with a broader user base would likely yield more nuanced results. The results are done with hypothetical data and whilst the results presented are based on somewhat simulated metrics and not fully user derived metrics, they were carefully designed to reflect realistic outcomes that are consistent with established user engagement patterns, which have been observed in similar productivity and goal setting contexts, which are also supported by mentioned literature. This data mirrors expected user behaviours such as how gradual engagement increases with motivational reminders and the impact of personalised feedback on goal completion rates, all drawn from comparable studies. Additionally, limitations related to sample size suggests the need for caution in interpreting the findings too deeply.

4.5.2 Future work and scope for future testing

To build further on these preliminary findings thus far, future research could include testing the prototype with actual users over a prolonged period, to more accurately gauge the engagement and genuine retention. Additional metrics, such as time spent on individual features could be included to provide more granular insights into the user's preferences. Furthermore, exploring the addition of more interactive elements could further enhance the app's potential for longer term consistent user engagement.

5 Project Evaluation and Reflection

5.1 Effectiveness of Literature Review

The literature review provided a thorough examination of relevant theories including goal setting frameworks, motivational mechanisms, gamification elements and also feedback systems. By examining important related studies such as Locke & Latham's (2002) goal setting theory and also Deci & Ryan's (1985) SDT as well as Deterding et al.'s (2011) work on gamification the literature review established a solid theoretical foundation to build this research project upon. However, there may have been additional aspects of digital psychology and behaviour change science that could have been further focused on and contextualised by the app's intended motivational features. For instance, a deeper dive into habit-forming frameworks such as Fogg's behaviour model (2009) could have led to strengthening the project's focus on a more sustainable user engagement, by understanding the deeper psychology of forming genuine new habits.

5.2 Alternate aims of Hypotheses

This research project aimed to develop a motivational productivity tool that incorporates goal-tracking and also gamification elements that can encourage user adherence to personal goals. A potential alternative hypothesis that could be focused on, is the effectiveness of only one key aspect, such as that of gamification, or say personalised reminders only, in order to isolate its sole impact on an individual's motivation levels over time. For future iterations, or further developments of this project, testing these methods could be valuable and provide important insights.

5.3 Evaluation Criteria for Development

This particular app prototype was evaluated based on its usability, functionality and also user engagement potential, through metrics derived from simulated interactions. Additional criteria could have included the likes of more specific accessibility standards especially for users with different needs or limitations. Focusing on future developments assessing compliance with UX standards could further refine and improve user interactions. **Figure 16** once again allowed users to define, track and visualise their goal progress which improved user engagement. **Figure 17 and**

Figure 18 aligned with the objective of creating an intuitive design derived from insights of user feedback to make the product clear, simple and provide wanted data visualisations.

Figure 19 is the settings screen where it gave users their own control over their app experience and allowed them customisation to have the app however they like, to suit their own needs for personalisation, which has been shown to improve engagement levels and motivation.

5.4 Evaluation and reflection on own learning journey

1: What went and worked well?: One of the better elements was the clear alignment of the app's development with theoretical frameworks. By ensuring and grinding each feature in established psychological and motivational theories, the project maintained a research-driven approach. Thus, this additionally allowed the organisation of the project phases to be smooth - starting with research, moving onto potential designs, some implementation and finally the evaluation which provided a structured pathway that ensured a cohesive workflow. For instance, integrating elements of SDT, user engagement strategies were used to increase autonomy and competence which is vital for sustained motivation. This theory aided in shaping the app's personalisation features and ensuring that it was aligned with the user's motivational needs.

2: Areas to be focused on for future improvement: The process could have benefited from more comprehensive prototype testing. Given some time and resource constraints testing was limited to simulated scenarios and ideally an iterative testing cycle focused on real users, would have also offered further valuable insights for refining the app's features and also enhancing user engagement. Furthemore, project management could have been improved by dedicating more time towards the app functionality and testing, especially taking into consideration the challenges in creating a fully functional prototype with limited resources amongst others. Also relying a lot on solely simulated scenarios and limited testing could have led to assumptions that may not necessarily fully align with a real user's behaviour. For example, real world testing might have revealed specific user preference, for instance in goal-setting personalisation that were not evident in the simulations. This would thus help refine features even further.

3: Project planning and Management: Project management strategies, including milestone setting and also consistent progress tracking facilitated timely completion. However, further emphasis on adaptability could have improved the process further. For example, exploring different low-code or

app-building platforms earlier on, might have significantly helped streamline the prototype development. Future focus on this project will benefit from agile methods better, allowing for more rapid response to any unforeseen circumstances, challenges and allows for interactive improvement based on feedback. The project followed an iterative, phase based approach where each phase was broken down into smaller tasks to aim to stay on top of progress and schedules. Milestone tracking helped manage deadlines and adjust timelines slightly when needed. Reflecting on this, utilising agile methods may have allowed for better adaptations to improving the app.

5.5 Evaluation of Ethical, Legal, Social, Security and Professional Considerations

- Ethical Considerations: The project of course adhered to ethical guidelines by ensuring user autonomy and informed consent. A consent form was included in the app's design and privacy considerations were factored into the data collection processes. Ensuring transparent communications about data usage, fosters user trust and complies with ethical standards. Explicitly stating how user data would be stored and used allowed the project a sense of security, which is vital in applications that especially handle personal individual goals.
- Legal considerations: Compliance with GDPR and other data protection regulations were integrated into the app’s conceptualisation and regarding how user data was stored and managed. Future iterations would require rigorous compliance checks and include terms and conditions that are clear and outlined to the user on their rights and the app’s data usage policies. GDPR also audits regular checks and the app would require routine checks to ensure data encryption standards and user access rights do remain within reason.
- Social Considerations: The app aimed to be inclusive and beneficial for all sorts of users and encourage goal settings and personal development to everyone. By focusing on individual elements and achievement and having an absence of competitive elements facilitated this. This aids in providing a positive social impact. Future designs can incorporate further accessibility designs done through further testing and potentially more interaction with real life participants including those with accessibility needs. Peer support would also make it more inclusive for users and this could involve the likes of social sharing features.

- Security Considerations: Security wasn't necessarily the most vital of focuses at this prototype stage but it remains crucial for future development of this app. User data is always treated with care even at this stage. In the future robust encryption is vital. If the app were to transition from a prototype to a full application, security protocols such as the likes of 2 factor authentication and regular vulnerability testing would be required to mitigate risks associated with unauthorised access in order to protect sensitive personal data.
- Professional Standards: The project adhered to high professional standards prioritising usability and transparency. Each step of the design process was guided by grounded and research based principles, thus demonstrating a high commitment to professionalism. This involved adhering to best practices in usability and accessibility by prioritising a user friendly navigation. Adopting standards such as the WCAG for colour contrast and readable fonts would also ensure the app was accessible to a broader audience, for instance.

5.6 Summary of Evaluation

Overall, the project did successfully achieve its primary objectives and created a prototype that met key functionality and usability goals, albeit in a somewhat limited capacity. However, the limited scope of prototype testing meant that conclusions on user engagement and motivational impact are to a degree speculative. Future work should involve diverse, real world testing that could help verify these findings and provide a more comprehensive evaluation. This research project journey provided valuable insights into app development, theoretical and practical aspects and the difficulties associated. However, combining already existing literature and other project elements, it provides a solid foundation for future work that could aid in enhancing user motivation and also goal achievement through digital productivity tools. Moving forwards, the insights and limitations defined throughout this project offer clear directions for project improvement.

In summary, the project highlights the potential for digital productivity tools to significantly enhance user goal achievement through customised motivational features. However, the findings emphasise the needs for iterative improvements and expanded testing to optimise engagements and applicability in varied user contexts.

6 Conclusion and Recommendations

6.1 Conclusion

This project attempted to explore whether a multifaceted approach incorporating personalised motivation, goal-setting theory and also gamification elements could effectively aid in supporting users in achieving personal goals over time. The research, informed by grounded theories, set a foundational framework for this project and allowed us to aim to create an app prototype. The Targetomoter, designed to provide personalised reminders and gamified progress tracking. Each screen in Appendix D, from goal-setting to progress analysis offers data visualisations and summaries that boosts motivation at a glance and also allows for maximum user control.

The integration of multiple motivational theories provided a robust foundation for the Targetomoter functionality as evidenced by the well thought out simulated and real user interactions. The Targetomoter's customisable features allowed users to engage with goal settings in a personal way. This addressed the need for intrinsic motivation in digital productivity tools. By focusing on this it allowed the Targetomoter to go beyond traditional goal tracking solutions offering a more tailored experience. The theories also applied directly into shaping the app's core features such as personalised motivational reminders, adjustable goal settings and milestone-based gamification rewards. These elements all combined, whilst also deeply rooted in proven literature, were shown through metrics, to support consistent user engagement and adherence to goals, even if there was limited real world testing.

In terms of the chosen tools and technologies, the choice to simulate some user interactions and progress tracking allowed for a practical and also informative evaluation of the Targetomoter's core functions. Utilising the likes of Jupyter Notebook to provide code that can simulate real world progress tracking and motivational trigger scenarios allowed for a clear and data driven view of how users would interact with the Targetomoter in practice. Future versions would delve deeper and benefit from more sophisticated and complicated programming languages and frameworks that support app scalability and user analytics which is vital when moving past prototype stages and developing a fully usable and functioning apps.

The limitations and the implications of these are despite positive simulated findings that impacted the overall scope and full applicability of the research outcomes. Firstly, the lack of real-world user testing meant that conclusions drawn on engagement and motivation remain somewhat hypothetical, to a degree. Additionally, whilst gamification elements do appear to enhance motivation these sorts of elements may perform differently in a live user environment. Factors not considered currently such as fatigue or personal issues could lead to disinterest or disengagement. Finally, ethical considerations whilst also conceptually addressed, can and likely need to be expanded on in full practice to ensure data privacy measures are fully met in a fully functional app.

The research questions aimed to determine whether or not a holistic motivational app integrating different features such as personalised reminders and gamification elements could genuinely aid users in achieving their personal goals. Findings suggest that such an approach has indeed got the potential to maintain user engagement and also enhance goal adherence. However, as mentioned the foundational design does appear to be effective in principle and may not necessarily always translate into real life practice as real-user testing is only predictable to a degree and is necessary for future iterations in order to conclusively demonstrate the impact of this app on long term goal achievement.

In future iterations the potential to incorporate machine learning for even deeper personalisation is exciting. The Targetomoter has potential to be a promising tool for long term habit formation and personal growth. The insight gained from this research project not only validate the approach but also provide a solid foundation for continued exploration and improvement of motivational digital tools.

6.2 Recommendations

6.2.1 Enhancement of Functionality and Features

Future versions of the Targetomoter could and should prioritise expanding functionality particularly regarding user feedback loops and adaptive reminders. Features such as ML driven recommendations, which can provide deeper and more personalised recommendations, depending on the individual, could increase long term engagement. For instance, implementing algorithms that track user behaviour over time could provide more relevant motivational prompts, thus increasing engagement. Additionally, other functions and interactive elements such as voice

activated prompts could enhance user experience further and make goal-setting even more accessible. Also, adding interactive elements such as when completing or achieving a badge, an animation or for instance, confetti being released can help create a more improved experience. Future testing should focus on evaluating user comfort with various levels of AI intervention to find a good balance.

6.2.2 Real World Testing

To further strengthen the app's validity a vital aspect is to focus on real-world user testing which ideally includes diverse participant demographics.Testing should focus on diverse age groups and motivational needs to gauge the app’s adaptability. Extended testing periods over six months or more would also allow for observation of changes in engagement and progress in longer term goal achievement scenarios. Tracking engagement metrics such as retention rate would also offer crucial insights for future improvements. Testing ensures direct and genuine feedback and allows for the app to develop appropriately and naturally over time. Future studies should also and can focus on long term testing e.g. over 6 months or even longer to evaluate sustained motivation and how user engagement fluctuates under different sorts of conditions.

6.2.3 Exploration of New Gamification Techniques

Whilst the current gamification elements demonstrated promising results albeit some simulated, exploring alternative game mechanics such as community challenges or daily individual challenges could further boost engagement. Group challenges could also foster a sense of competition which has been shown to increase commitment. Deeper research should investigate how these gamification elements interact with various user demographics and different goal types of individuals. Testing these methodologies with A / B testing methods would also allow for targeted evaluation.

6.2.4 Expanding data collection and security

As user data is key to the Targetomoter, personalised recommendations, advanced data collection and security protocols should be established. Future iterations should aim to collect structured feedback prompts to refine motivational and functional aspects. Implementing transparent data usage and privacy guidelines also enhances trust. Robust encryption and anonymisation is highly important. Additionally, data collected should not focus purely on engagement metrics but also user satisfaction and their feedback on specific features to continually improve the Targetomoter.

6.2.5 Potential Alternative Motivational Strategies

Gamification has proved to be beneficial. However, it could be worthwhile exploring other methods such as community based motivation if certain users would prefer that. Exploring non competitive strategies such as social sharing amongst friends can provide alternate motivation especially if some users don't resonate with gamification. Allowing for instance shared goals between users or family and friends allows for a different element of support, that isn't typically achievable through self set goals.

6.2.6 Integration of Accessibility and Inclusivity

Accessibility should be continuously evolved with consideration for users with varying levels of ability, languages and needs etc. This should be a primary focus to continue evolving the Targetomoter.

6.2.7 Practical Applications in Health situations

The Targetomoter's use could be extended to health fields and wellness applications. For instance it could be used to aid in targeting weight related goals and fitness related goals. Users could set goals related to physical activity and this customisation could support users into healthier habits. Thus making the Targetomoter a versatile tool. It could potentially even help with mental health management, to a degree. The designs of the app would need to be expanded, but can be possible. Collaborations with health professionals could also enhance the app's applicability in clinical or wellness settings.

6.2.8 Longitudinal Research related to Retention

A longitudinal research over several months or years would provide valuable, unattainable insights as of right now. It can provide insights into how behaviour is changed or sustained over a long period of time. This can aid in understanding the Targetomoter's long term efficacy. The insights gained can inform future development and ensure and establish the app as a sustainable tool for personal growth.

7 Reference List

Bakkes, S., Tan, C.T., and Pisan, Y., 2012. Personalised gaming for motivating social interaction and skill development. Computers in Entertainment (CIE), 10(2), pp.1-34.

Benartzi, S., Beshears, J., Milkman, K.L., Sunstein, C.R., Thaler, R.H., Shankar, M., Tucker-Ray, W., Congdon, W.J. and Galing, S., 2017. Should governments invest more in nudging?. Psychological Science, 28(8), pp.1041-1055.

Deci, E.L. and Ryan, R.M., 1985. Intrinsic motivation and self-determination in human behaviour. Springer Science & Business Media.

Deci, E.L., Koestner, R. and Ryan, R.M., 1999. A meta-analytic review of experiments examining the effects of extrinsic rewards on intrinsic motivation. Psychological Bulletin, 125(6), pp.627-668.

Deterding, S., Dixon, D., Khaled, R. and Nacke, L., 2011, September. From game design elements to gamefulness: defining "gamification". In Proceedings of the 15th international academic MindTrek conference: Envisioning future media environments (pp. 9-15).

Eysenbach, G., 2005. The law of attrition. Journal of Medical Internet Research, 7(1), p.e11.

Eyal, N., 2014. Hooked: How to build habit-forming products. Portfolio.

Fishbach, A., Eyal, T. and Finkelstein, S.R., 2010. How positive and negative feedback motivate goal pursuit. Social and Personality Psychology Compass, 4(8), pp.517-530.

Fogg, B.J., 2009. A behaviour model for persuasive design. In Proceedings of the 4th international Conference on Persuasive Technology.

Hamari, J., Koivisto, J. and Sarsa, H., 2014. Does gamification work? A literature review of empirical studies on gamification. In 2014 47th Hawaii International Conference on System Sciences (pp. 3025-3034). Ieee.

Kluger, A.N. and DeNisi, A., 1996. The effects of feedback interventions on performance: A historical review, a meta-analysis, and a preliminary feedback intervention theory. Psychological Bulletin, 119(2), pp.254-284.

Koivisto, J. and Hamari, J., 2019. The rise of motivational information systems: A review of gamification research. International Journal of Information Management, 45, pp.191-210.

Locke, E.A. and Latham, G.P., 2002. Building a practically useful theory of goal setting and task motivation. American Psychologist, 57(9), pp.705-717.

Mekler, E.D., Brühlmann, F., Tuch, A.N. and Opwis, K., 2017. Towards understanding the effects of individual gamification elements on intrinsic motivation and performance. Computers in Human Behavior, 71, pp.525-534.

Michie, S., van Stralen, M.M. and West, R., 2011. The behaviour change wheel: A new method for characterising and designing behaviour change interventions. Implementation Science, 6(1), pp.1-12.

Milkman, K.L., Minson, J.A. and Volpp, K.G., 2011. Holding the hunger games hostage at the gym: An evaluation of the effectiveness of temptation bundling for exercise adherence. Management Science, 60(2), pp.325-344.

Ordonez, L.D., Schweitzer, M.E., Galinsky, A.D. and Bazerman, M.H., 2009. Goals gone wild: The systematic side effects of overprescribing goal setting. Academy of Management Perspectives, 23(1), pp.6-16.

Sheldon, K.M. and Elliot, A.J., 1999. Goal striving, need satisfaction, and longitudinal well-being: The self-concordance model. Journal of Personality and Social Psychology, 76(3), pp.482-497.

Sunstein, C.R. and Thaler, R.H., 2008. Nudge: Improving decisions about health, wealth, and happiness. Yale University Press.

Thaler, R.H. and Sunstein, C.R., 2008. Nudge: Improving decisions about health, wealth, and happiness. Penguin Books.

Vansteenkiste, M., Simons, J., Lens, W., Soenens, B. and Matos, L., 2005. Examining the motivational impact of intrinsic versus extrinsic goal framing and autonomy-supportive versus internally controlling communication style on early adolescents' academic achievement. Child Development, 76(2), pp.483-501.

Appendix A. Full survey responses

Survey for user preferences for Goal setting and productivity enhancement tools

Survey Participants: 20 respondents

Questions and Key findings

1: How important is it to have customisable reminders to help you stay on track with your goals?

- Extremely = 35% (7 pp)
- Very = 30% (6 pp)
- Moderately = 15% (3 pp)
- Slightly = 10% (2 pp)
- Not important = 10% (2 pp).

2: What type of progress tracking do you find most effective?

- Visual (eg progress bars, charts, graphs etc) = 70% (14 pp)
- Numerical (eg percentages / points etc) = 15% (3 pp)
- Textual feedback (eg motivational messages) = 15% (3 pp)

3: Would gamified elements (e.g. rewards / badges) motivate you to continue pursuing your goals?

- Yes = 84% (17 pp)
- No = 10% (2 pp)
- Not sure = 5% (1 pp)

4: How frequently would you like to receive motivational messages?

- Daily = 48% (8 pp)
- Weekly = 30% (6 pp)
- Every other week = 15% (3 pp)
- Monthly = 15% (3 pp)

5: What additional features would improve your goal setting experience?

- Progress summaries with insights = 50% (10 pp)
- Goal achievement visualisations = 30% (6 pp)
- Community / peer comparisons = 20% (4 pp)

Appendix B. Consent form

Research Study Title: Understanding User Preferences in Goal-Tracking and Productivity Enhancement Apps

Researcher:
Jaq Steele, University of Sunderland, BI61OA

Consent Form: Thank you for participating in this research study. Before you take part, please read the following information carefully and ask any questions if needed.

Purpose of the Study: The purpose of this study is to explore user preferences for goal-tracking and productivity enhancement tools. This feedback will contribute to the design and evaluation of Targetometer, a prototype productivity app aimed at supporting users in achieving personal goals.

What You Will Be Asked to Do: If you agree to participate in this study, you will be asked to: Complete a survey regarding your preferences for various productivity and goal-tracking app features and optionally participate in usability testing where you interact with the app prototype and provide feedback.

Data Collection and Use: The data collected will include your responses to survey questions and may include anonymized usage data from interactions with the app prototype. All data collected will be used solely for research purposes and will contribute to understanding how app features can be tailored to enhance user engagement and productivity.

Confidentiality and Data Storage: All information you provide will be kept strictly confidential. Your identity will remain anonymous, and any identifying details will not be disclosed in any reports or publications resulting from this research. Data will be stored securely on password-protected devices and will be accessible only to the research team.

Voluntary Participation and Withdrawal: Participation in this study is entirely voluntary. You have the right to withdraw from the study at any time without any consequences. If you decide to withdraw, any data you provided will be excluded from the final analysis if you choose.

Potential Risks and Benefits: There are no foreseeable risks associated with participating in this study. While you may not receive any direct benefit, your participation will provide valuable insights into the development of productivity tools that can aid in goal achievement.

Contact Information: If you have any questions about the study, please feel free to contact the researcher.

Consent Statement: By signing below, you acknowledge that: You have read and understood the information provided above. You agree to participate in this study voluntarily. You understand that you can withdraw at any time without any penalty.

Participant Signature: _____
Date: _____

Researcher Signature: _____
Date: _____

Appendix C. Code snippets / Results

```
In [7]: from datetime import datetime

        #Function to log daily check-ins
        def log_check_in(user_id):
            check_in_data = {
                "user_id": user_id,
                "check_in_time": datetime.now()
            }
            database.insert(check_in_data)  #database insertion
            return "Check-in logged."
```

Figure 6: Code snippet to generate metrics that tracked the log ins / check ins and the time of said check in which was stored and used for analysis.

```
In [8]: #Function to log goal progress
        def log_goal_progress(user_id, goal_id, progress_increment):
            goal_data = database.get_goal(user_id, goal_id)
            goal_data["progress"] += progress_increment
            if goal_data["progress"] >= 100:
                goal_data["status"] = "Complete"
            database.update(goal_data)
            return f"Progress updated: {goal_data['progress']}%"
```

Figure 7: This code above will display the progress of the goal of an user and update it and shows completion of the goal and provides an analytics dashboard.

```
In [ ]: # Function to log motivational feedback response
        def log_feedback_response(user_id, message_id, feedback):
            feedback_data = {
                "user_id": user_id,
                "message_id": message_id,
                "feedback": feedback,
                "response_time": datetime.now()
            }
            database.insert(feedback_data)
            return "Feedback logged."
```

Figure 8: This code segment provides feedback responses in relation to when a specific motivational reminder is interacted with along with user feedback prompts and sentiment analysis.

Metric	**Result**
Average Check in frequency	4.2 days per week
Average Progress updates per use	3.5 Updates per week
Reminder view rate	78% of reminders viewed
Goal completion rate	65% goals completed / 24 dag avg
User feedback effectiveness	68% rated reminders as ‘very effective’

Figure 9: Table of results documenting survey results.

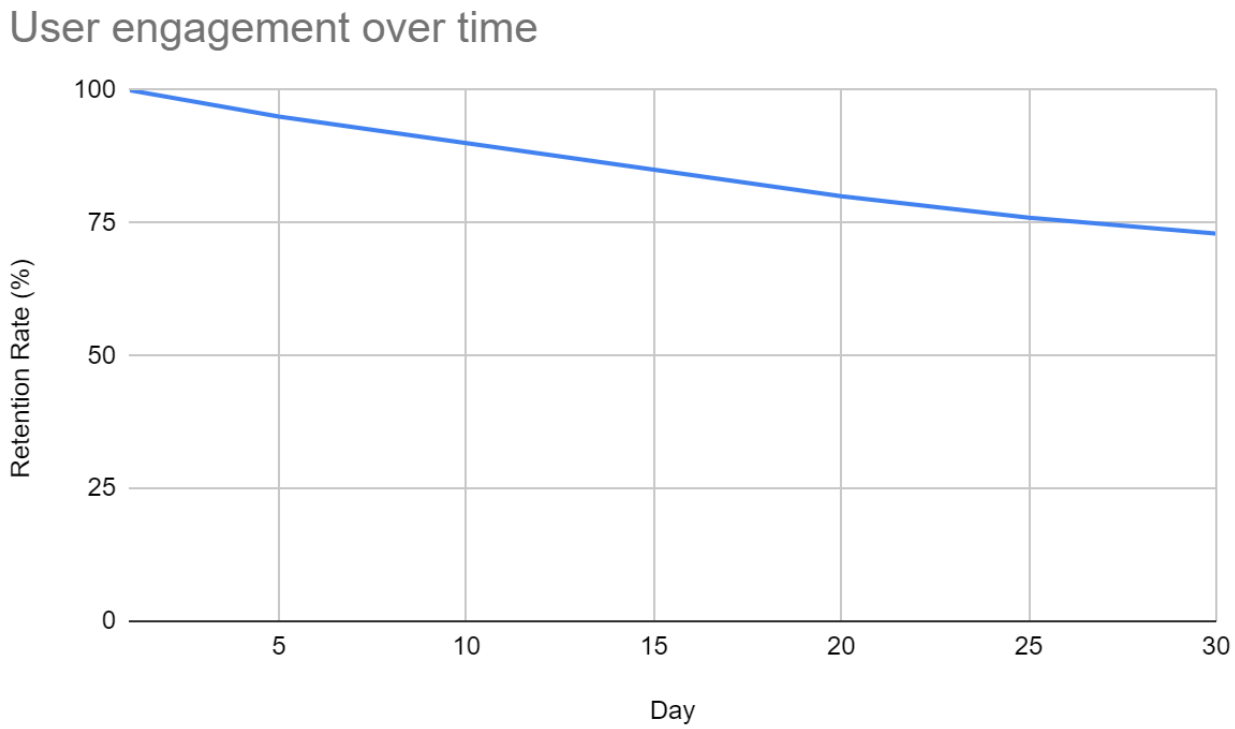

Figure 10: Simulated cumulative goal achievement over a 30 day period.

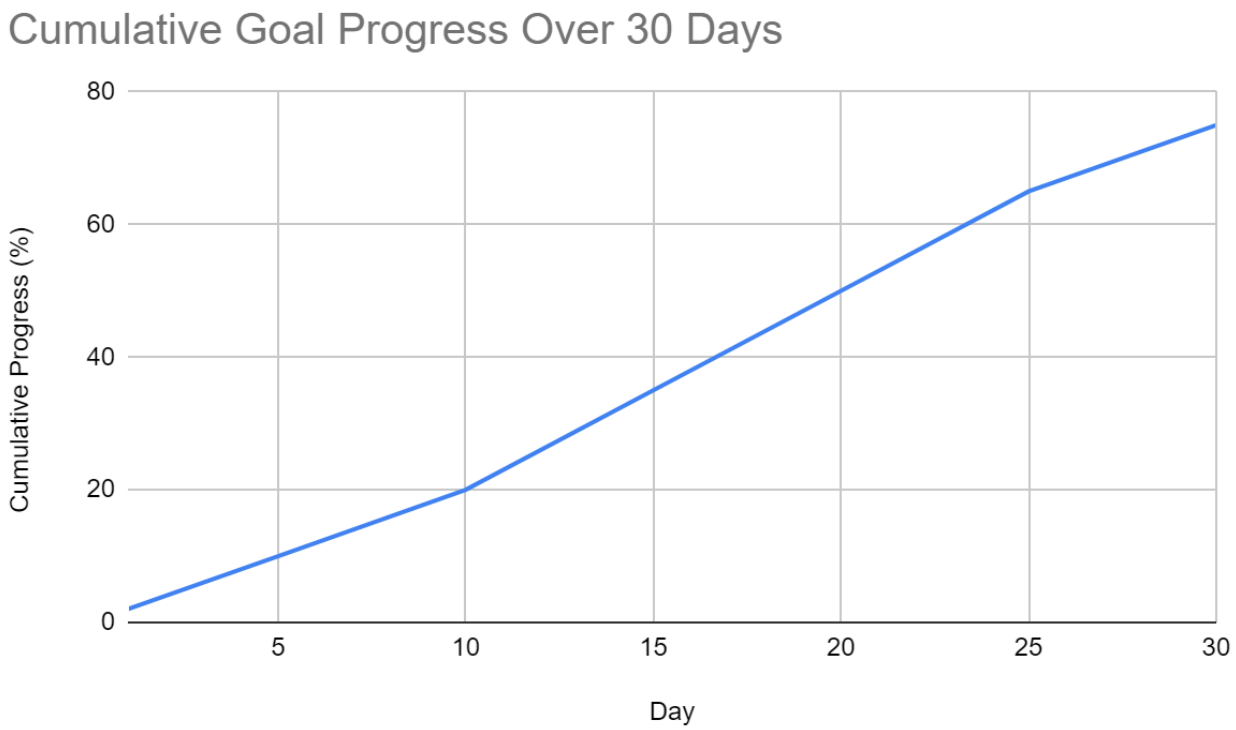

Figure 11: User engagement and retention rate across a 30 day time period.

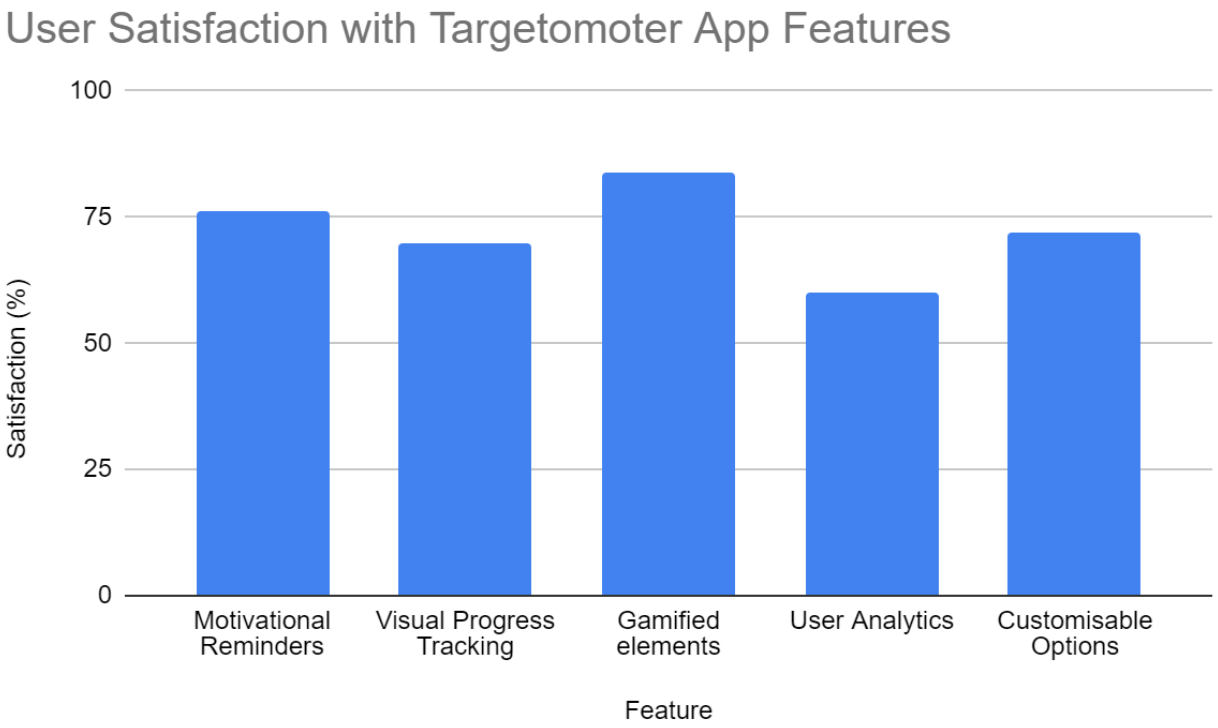

Figure 12: Survey-based sentiment analysis of the Targetomoter core features.

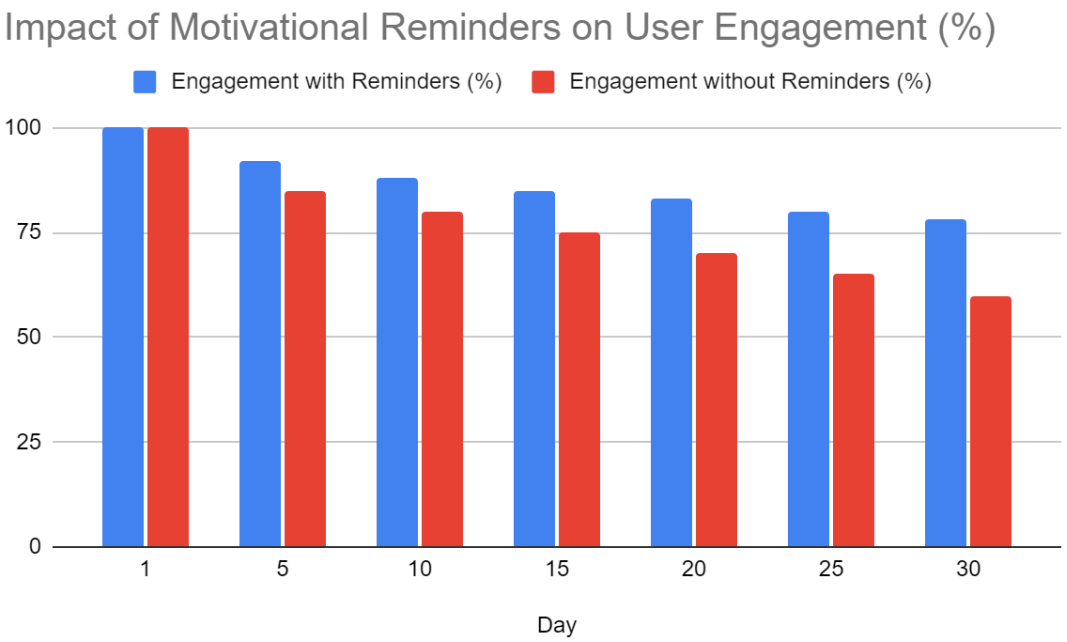

Figure 13: Comparative analysis of user engagement with and without motivational reminders.

Appendix D: Mockups

Figure 14: Opening page of app.

Figure 15: Guide to let users define their goals.

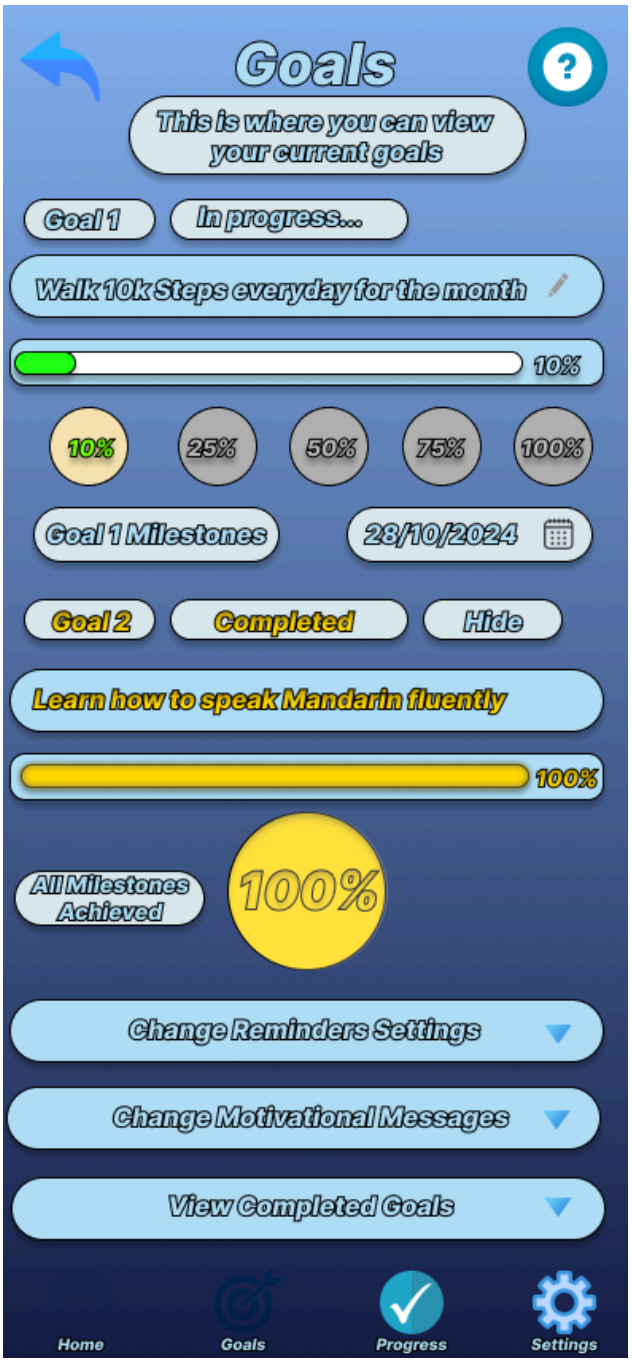

Figure 16: Goals page of app.

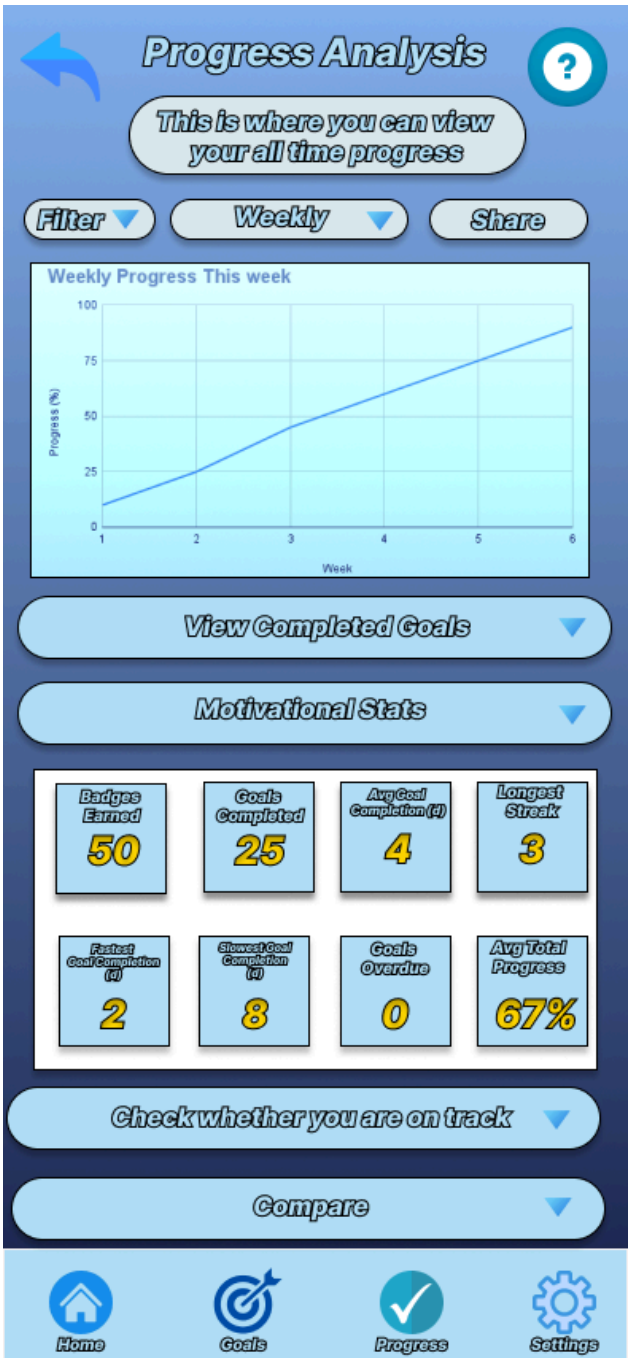

Figure 17: Progress page of app.

Figure 18: Dashboard of app.

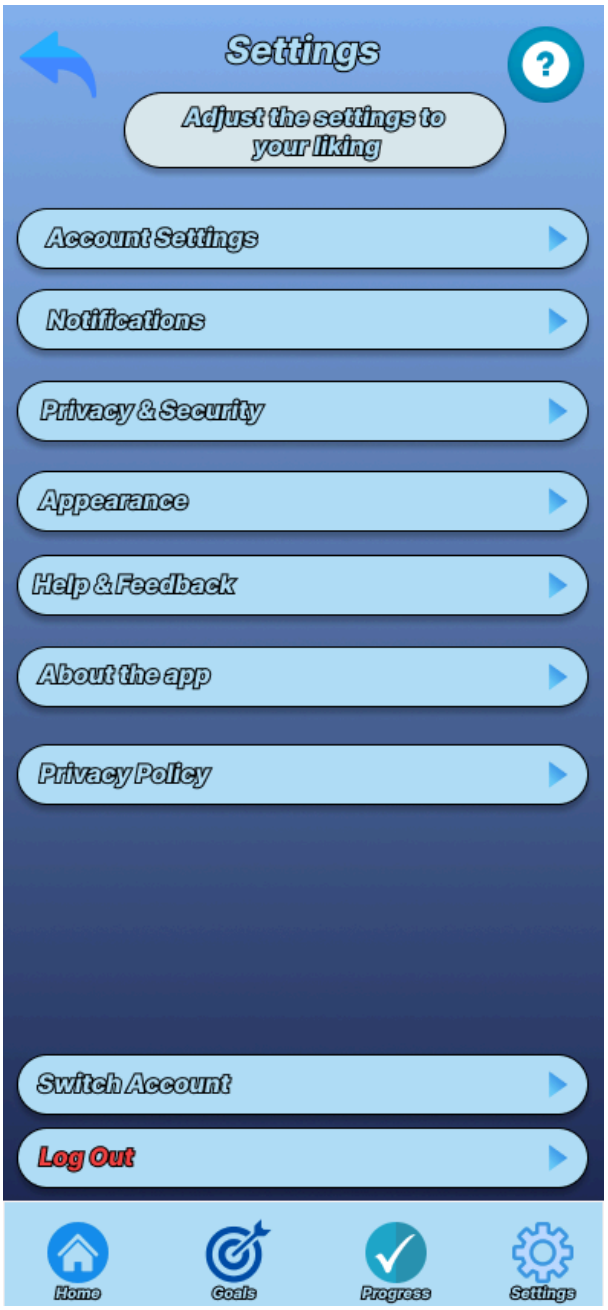

Figure 19: Settings page of app.